# 2.4 GHz FMCW Radar — Simulation Results

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The horn antenna was simulated in Ansys HFSS (Student 2025 R2) and the baseband board in NI Multisim 14.2 using library parts. Nothing has been built yet. Plots for every check are in Section 4.

## 1 Horn antenna (HFSS)

Pyramidal horn on WR-340, mouth 263.8 × 193.1 mm, coax probe feed. Band: 2400–2483.5 MHz.

| Check | Result | Requirement | |
|---|---|---|---|
| Match (S11), worst in band, original probe (28 / 43.7 mm) | −5.2 dB | ≤ −10 dB | **FAIL** |
| Match (S11), worst in band, final probe (27 / 17.7 mm) | −24.3 dB | ≤ −10 dB | **PASS** |
| Gain at 2.400 / 2.442 / 2.483 GHz | 13.24 / 13.24 / 13.27 dBi | 12.5–14 dBi | **PASS** |
| Beamwidth, E / H plane | 34.4° / 36.4° | theory 33.4° / 35.2° | **PASS** |
| TX → RX leakage, horns 290 mm apart | −49.4 dB | ≤ −35 dB | **PASS** |
| Match with probe off by ±1 mm | −14.5 dB | ≤ −10 dB | **PASS** |
| Mesh check (8.8k / 11.9k / 28.8k cells) | 0.14 dB gain spread | ≤ 0.2 dB | **PASS** |
| Copper walls vs. perfect conductor | +0.01 dB gain | ≤ 0.2 dB | **PASS** |

**Change made:** probe moved from 28 mm long, 43.7 mm from the back wall, to 27 mm long, 17.7 mm from the back wall (picked from a 1 mm sweep of 57 runs).

## 2 Baseband board (Multisim)

| Card | Check | Result | Requirement | |
|---|---|---|---|---|
| 1 Video amp | Outputs at half rail (DC) | 1.07 mV off | $\leq$ 150 mV | **PASS** |
| | Gain at 159 Hz / 1 kHz / 15.9 kHz | 48.7 / 54.5 / 50.9 dB | 48.7 / 54.5 / 50.6 dB | **PASS** |
| | 0.5 mV, 1 kHz in | 0.266 V out | 0.27 V | **PASS** |
| | First power-up gain, 10 mV in | 106.8 mV out | 107 mV | **PASS** |
| | Clipping | $\pm$4.06 V | about $\pm$4.2 V | **PASS** |
| | Two-tone: 417 / 870 Hz lines | 19.1 / 21.1 mV | 19.0 / 21.2 mV | **PASS** |
| 2 Half-rail ref | Divider / buffered output | 5.70 / 5.58 V | 5.70 / 5.58–5.72 V | **PASS** |
| | 150 kHz ripple reaching vref | $-$162 dB | $\leq -100$ dB | **PASS** |
| 3 12 V input | Protected / analog rail | 11.72 / 11.62 V | 11.68 / 11.58 V | **PASS** |
| | Ripple on analog rail | 3.2 mVpp | about 4 mVpp | **PASS** |
| 4 Reverse supply | $-$12 V applied | $-$11 mV gets through | $<$ 0.5 V | **PASS** |
| 5 Sync divider | Sync pulse level | 275 mV | 270–300 mV | **PASS** |
| 6 IF input | At 417 Hz / 2.44 GHz | $-$6.6 / $-$57.7 dB | $-$6.6 / $-$57.7 dB | **PASS** |
| 7 Scaled radar | Beat tone at 5 / 10 / 15 m | 435.0 / 870.7 / 1305.8 Hz | 435.2 / 870.4 / 1305.6 Hz | **PASS** |
| | Beat tone with leakage on | 872.7 Hz | stays at 870 Hz | **PASS** |
| 8 Full board (as built) | Outputs at half rail (DC) | 4.36 V off | $\leq$ 150 mV | **FAIL** |
| 8 Full board (fixed) | Outputs at half rail (DC) | 1.08 mV off | $\leq$ 150 mV | **PASS** |
| | Gain vs. card 1 (C6 = 47 µF) | 6.4 dB low | within 0.3 dB | **FAIL** |
| | Gain vs. card 1 (C6 = 3.3 mF) | 0.15 dB | within 0.3 dB | **PASS** |
| | Two-tone 870 Hz line | 21.0 mV | 21.2 mV | **PASS** |
| | 150 kHz ripple at output | 0.06 µV | $<$ 1 mV | **PASS** |
| | Power-up from 6 V | settles in 0.53 s | $<$ 1 s | **PASS** |

**Changes made on card 8:**

- Wired U1A's $V-$ and U1B's $V+$ pins. They were left unconnected when the blocks were copied onto one sheet, which latched the amp at the rail.
- C6 raised from 47 µF to 3.3 mF. The half-rail reference was moving with the signal and the first stage amplified it.

**Notes:**

- Card 8's analog rail reads 11.49 V because the TI op-amp model draws 5.7 mA per amp (datasheet: 1.4 mA typ.). Expected about 11.6 V on the real board.
- The op-amp model can't run below about 4.4 V supply, so power-up starts at 6 V.

## 3 Summary

- Horn: all checks pass after the probe move.
- Baseband: all cards pass. Card 8 passes after the wiring fix and the C6 change.

# 4 Figures

## Horn antenna (HFSS)

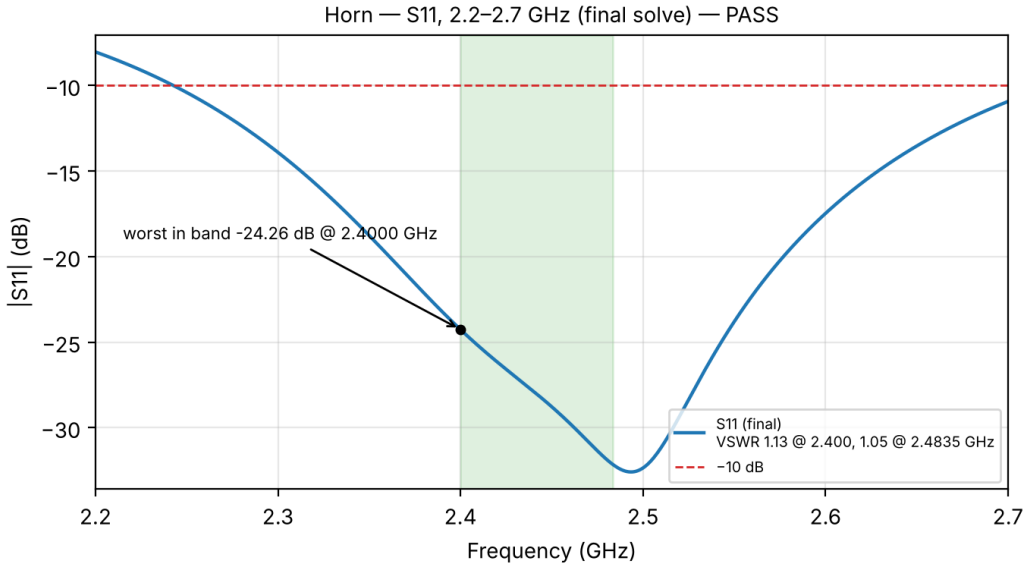



Figure 1: Match (S11) across the band, final probe

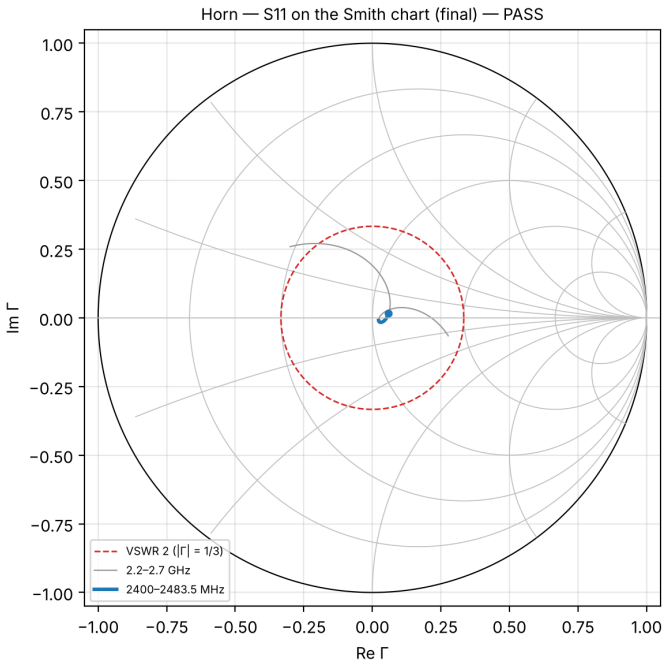



Figure 2: Smith chart, final probe

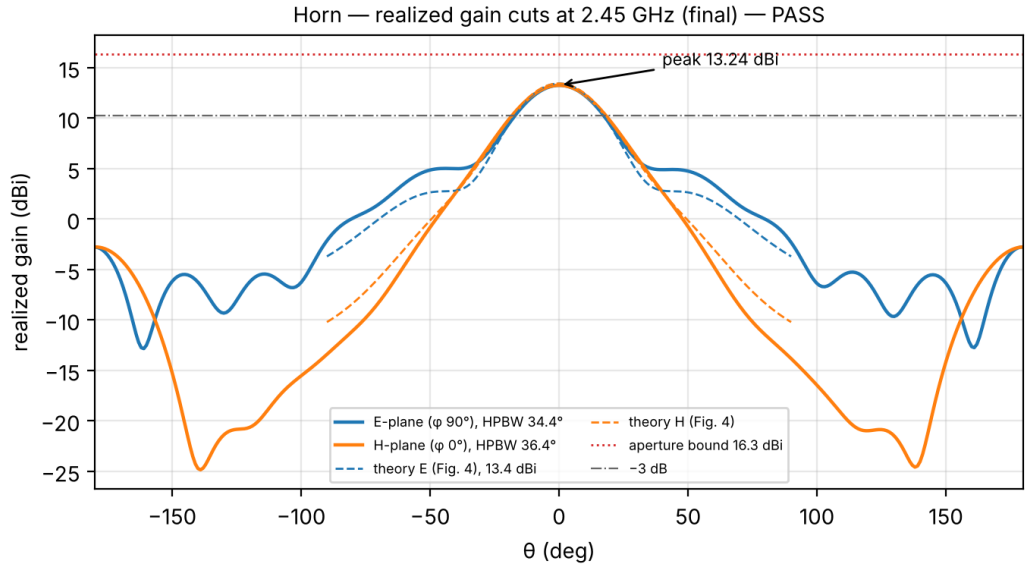



Figure 3: Gain patterns, E and H plane

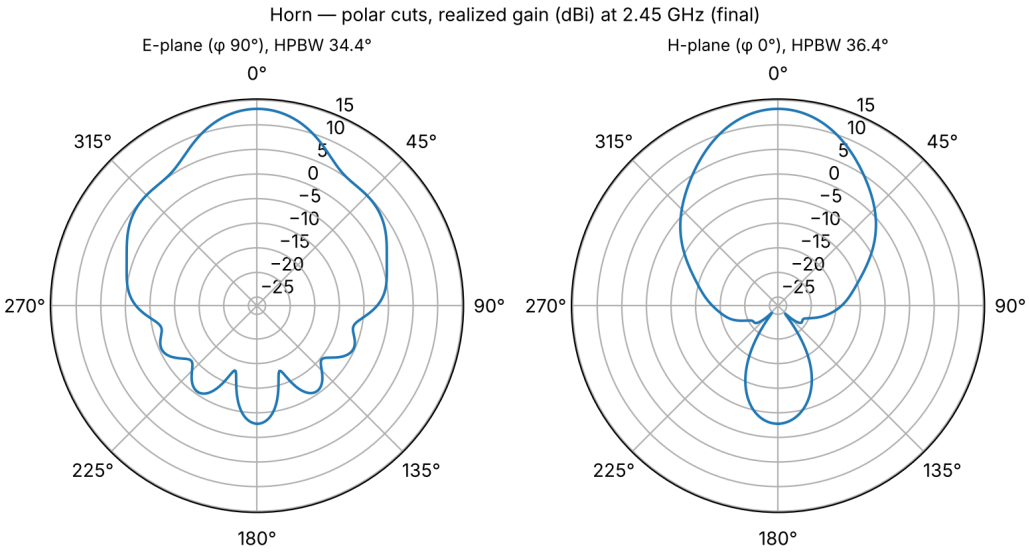



Figure 4: Gain patterns, polar

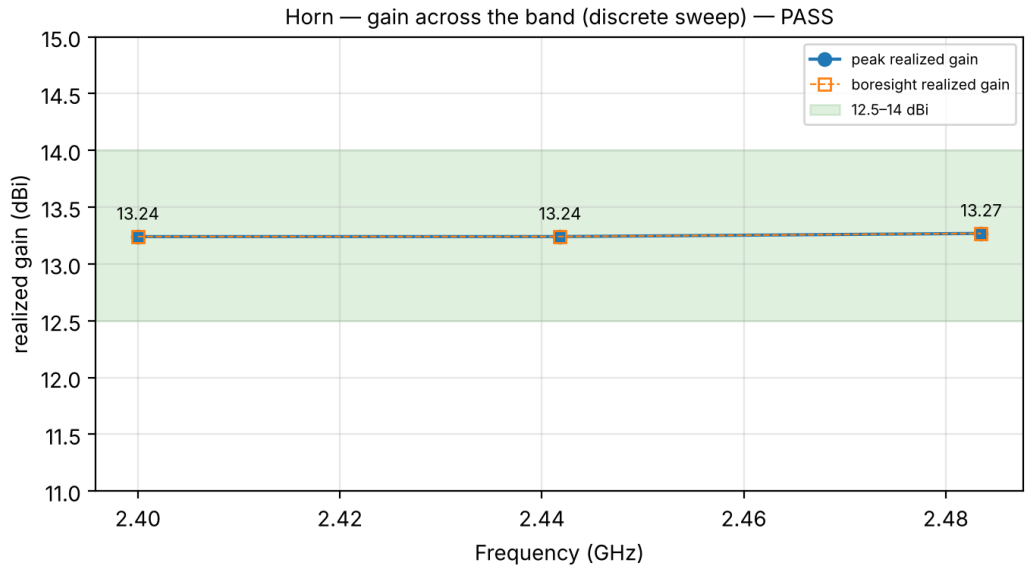



Figure 5: Gain straight ahead across the band

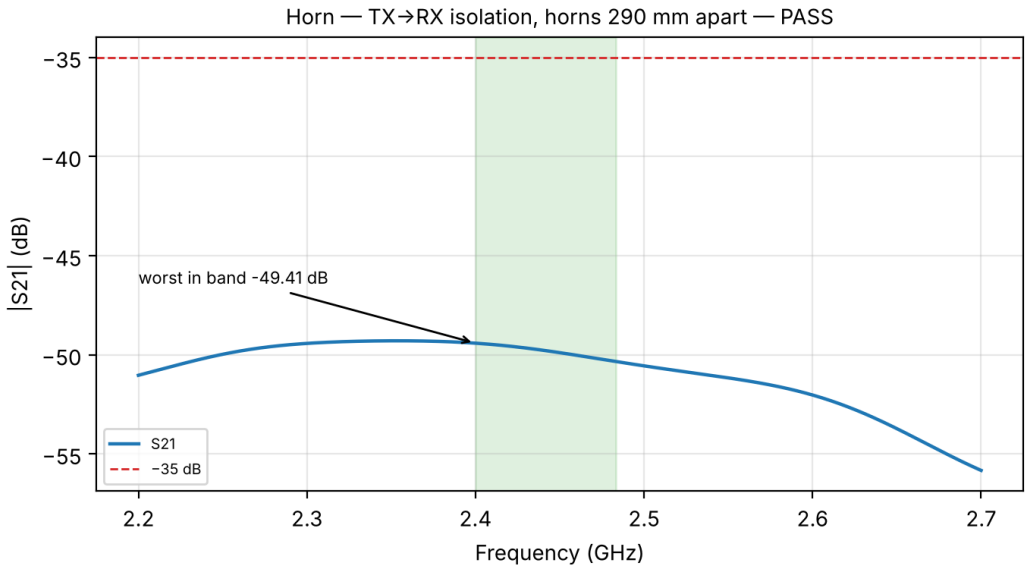



Figure 6: TX to RX leakage, horns 290 mm apart

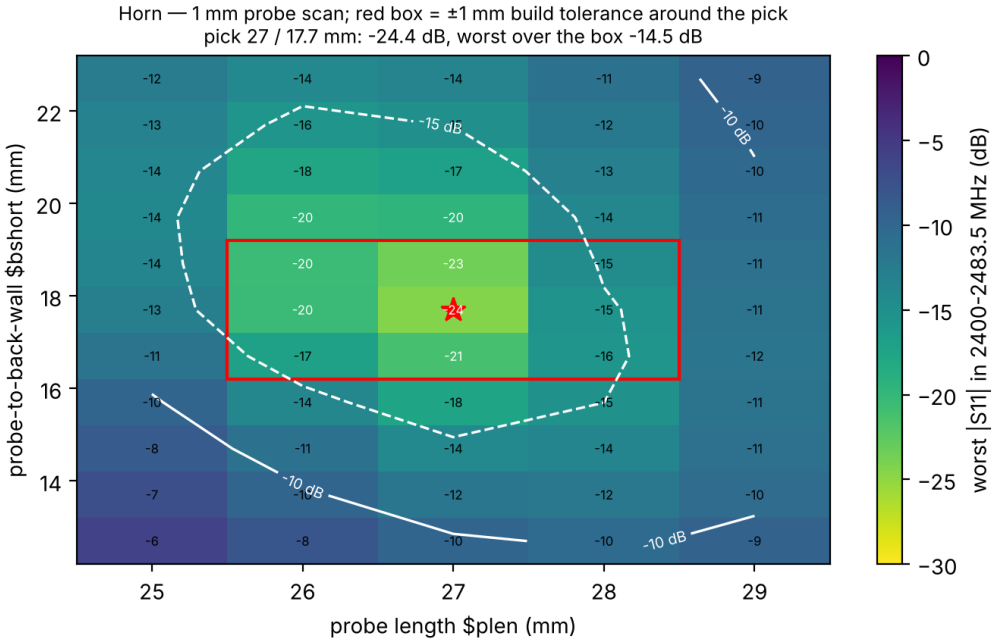



Figure 7: 1 mm probe sweep: worst match in band

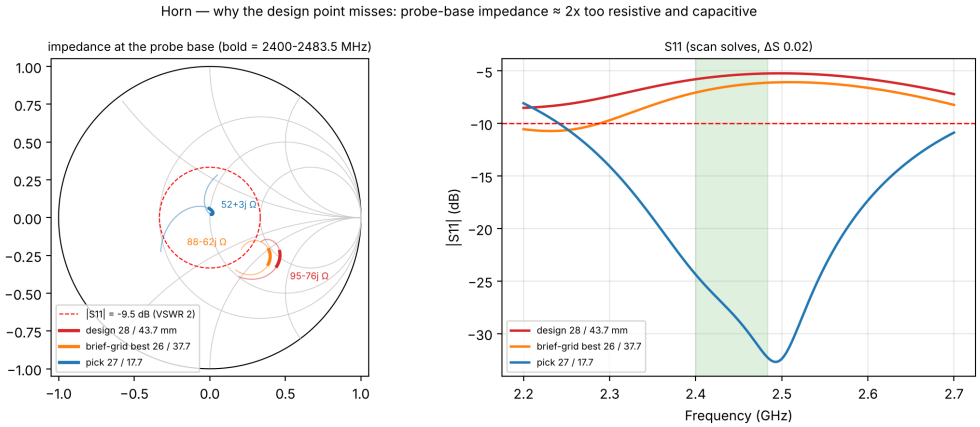



Figure 8: Why the original probe position didn't match

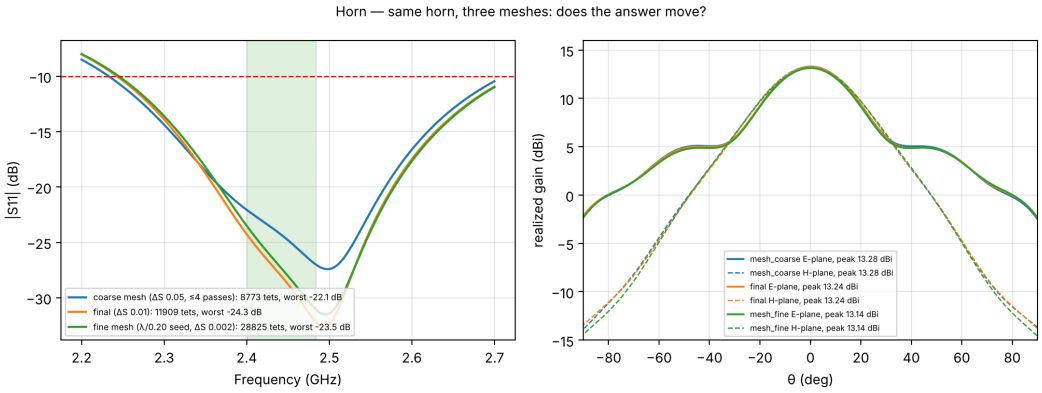



Figure 9: Mesh check

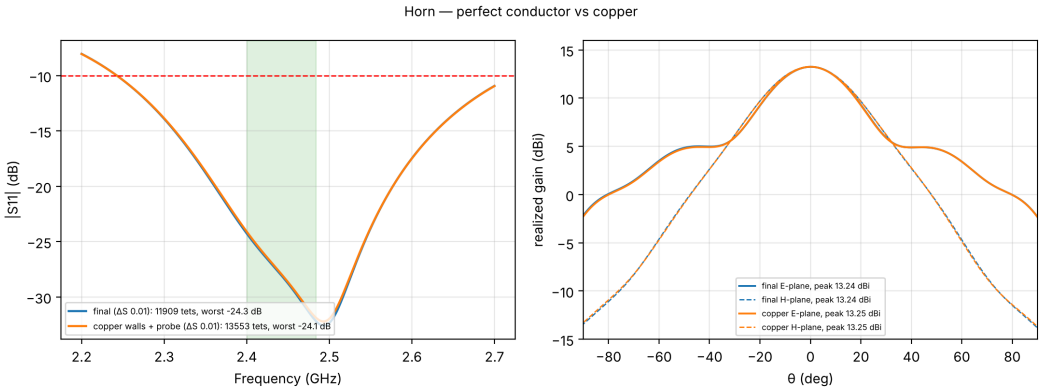



Figure 10: Copper vs. perfect conductor

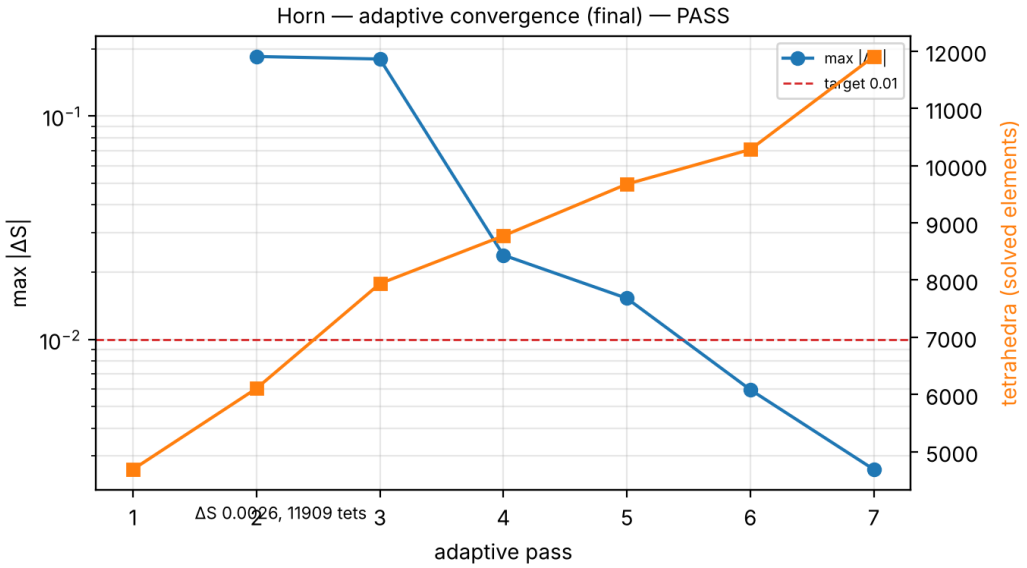



Figure 11: Mesh refinement

## Baseband board (Multisim)

### Card 1: video amp

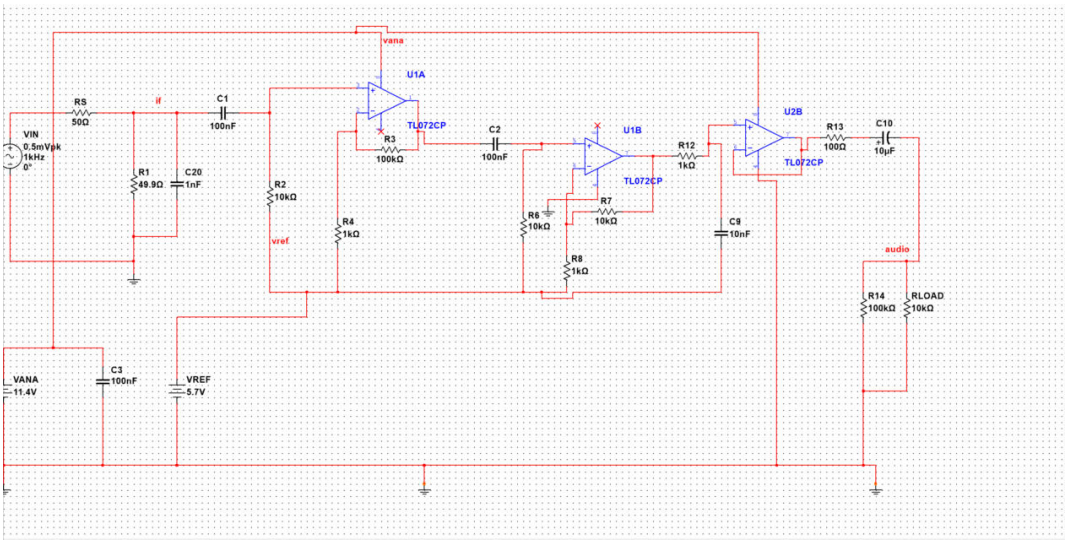



Figure 12: Multisim schematic

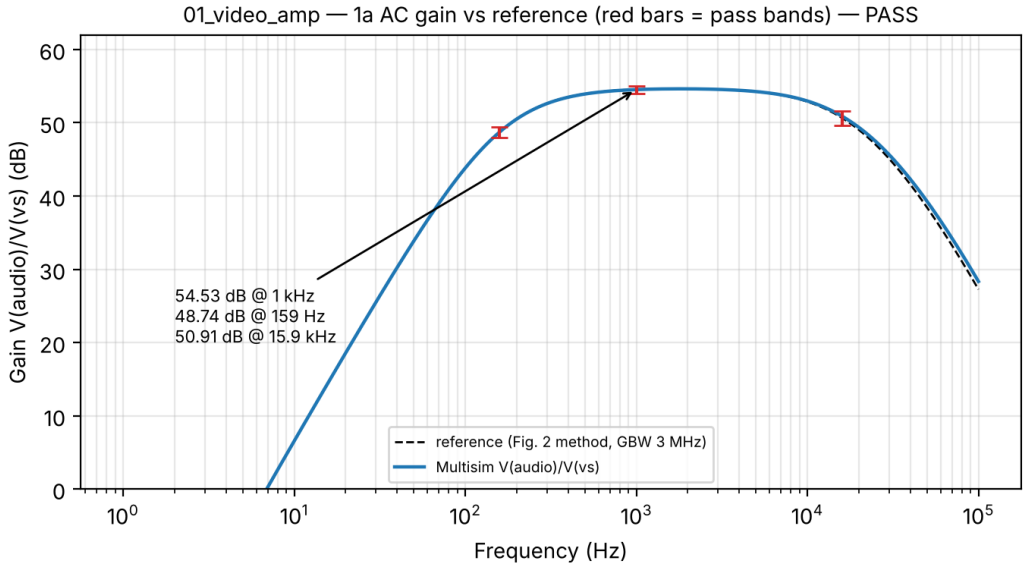



Figure 13: Gain vs. frequency

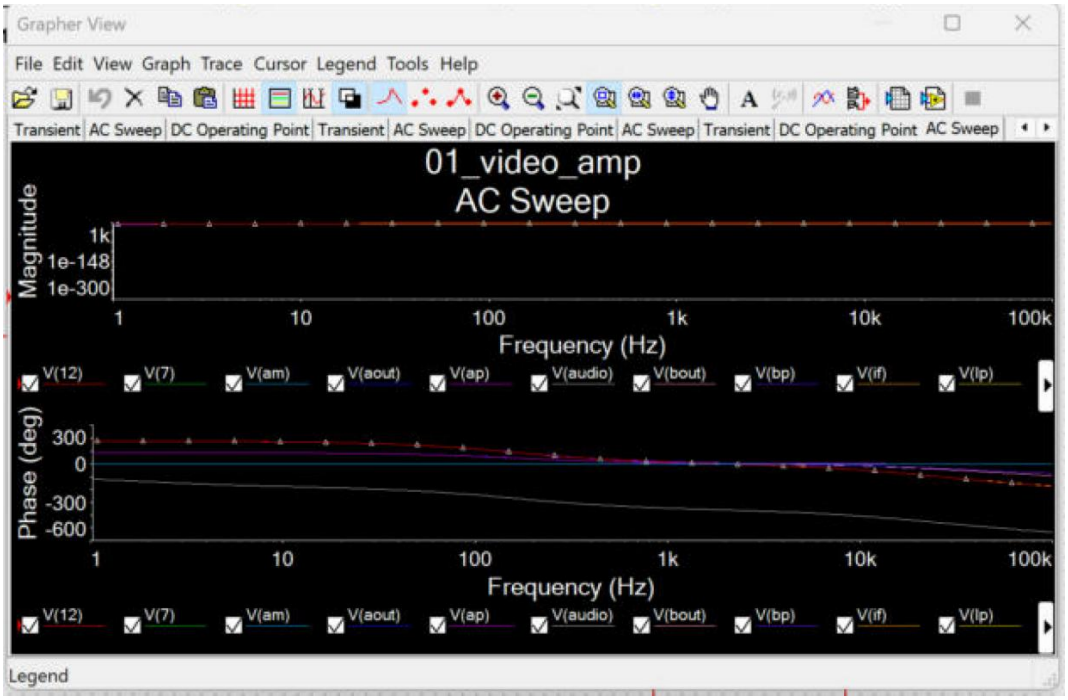



Figure 14: Multisim grapher, AC sweep

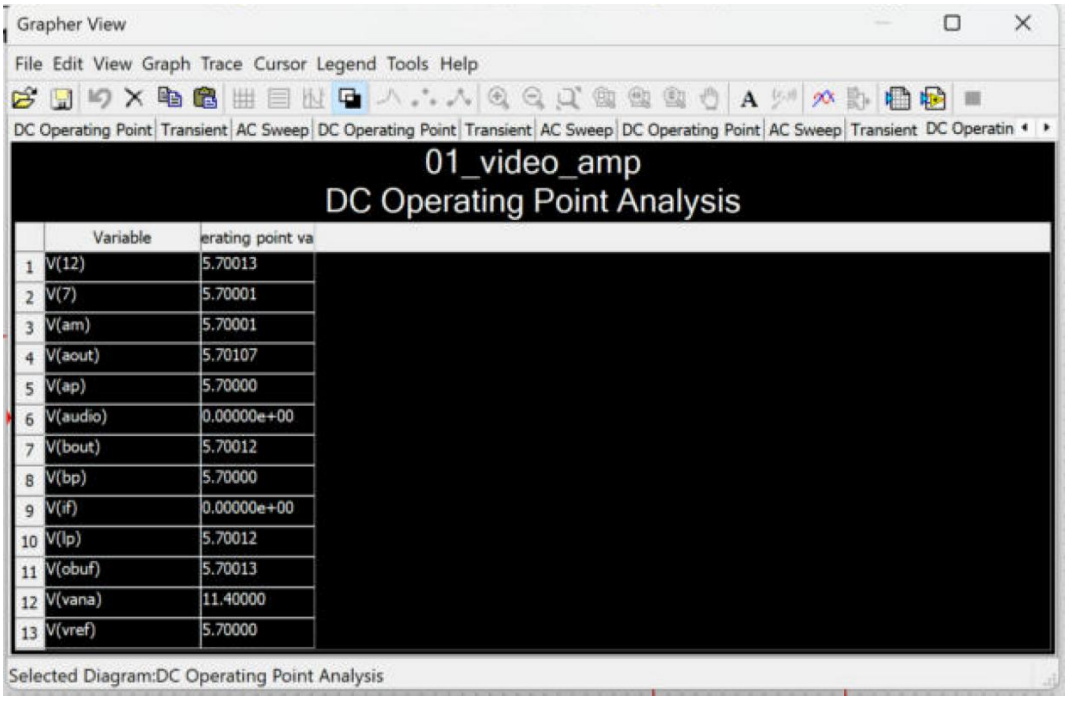



Figure 15: Multisim grapher, DC operating point

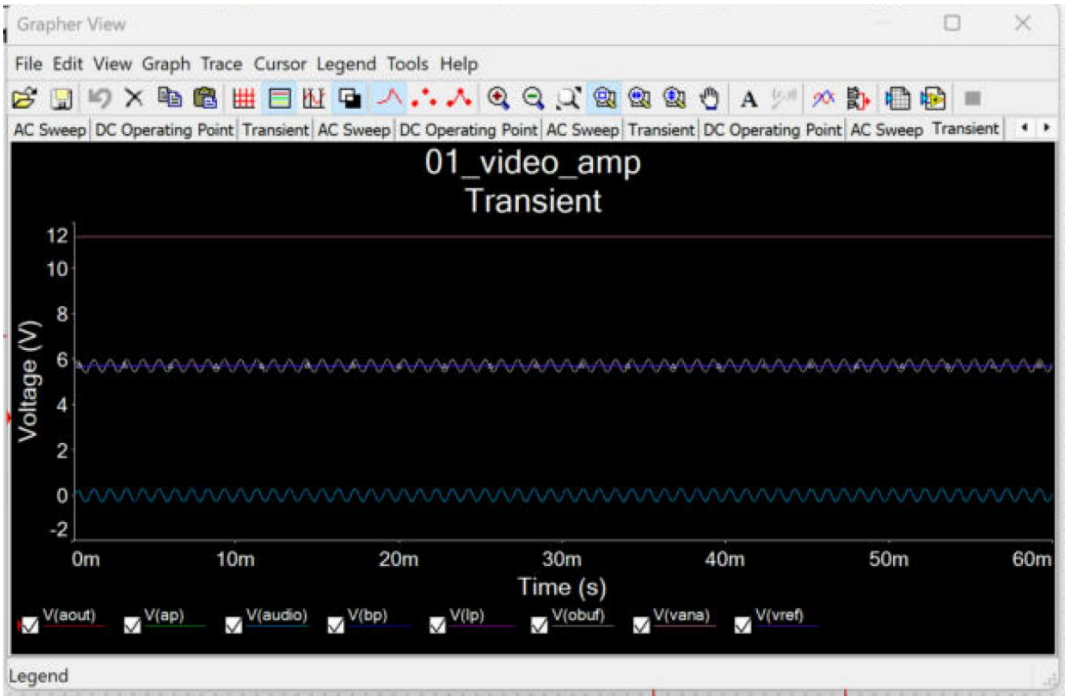



Figure 16: Multisim grapher, transient

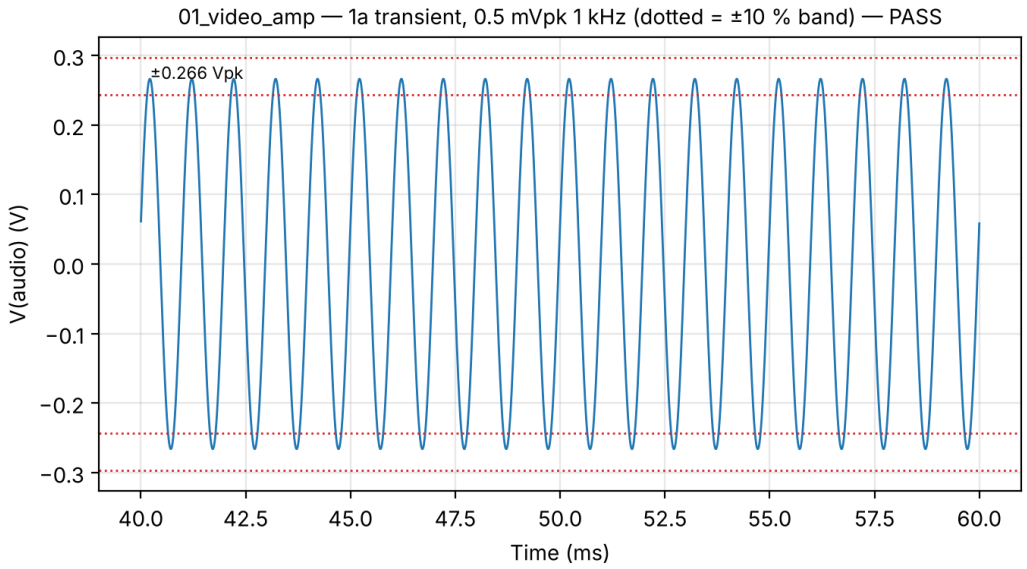



Figure 17: 1 kHz transient

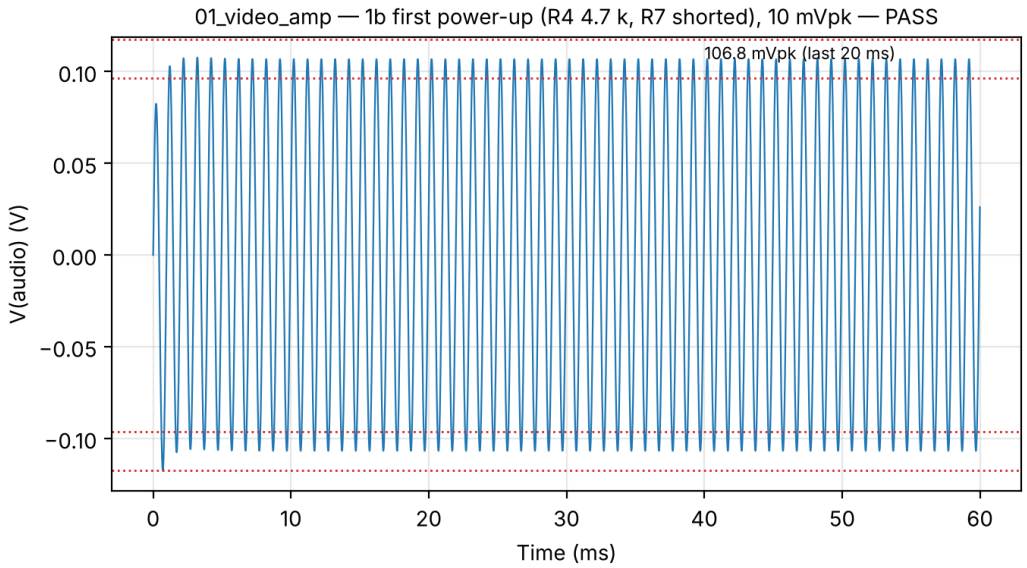



Figure 18: First power-up gain

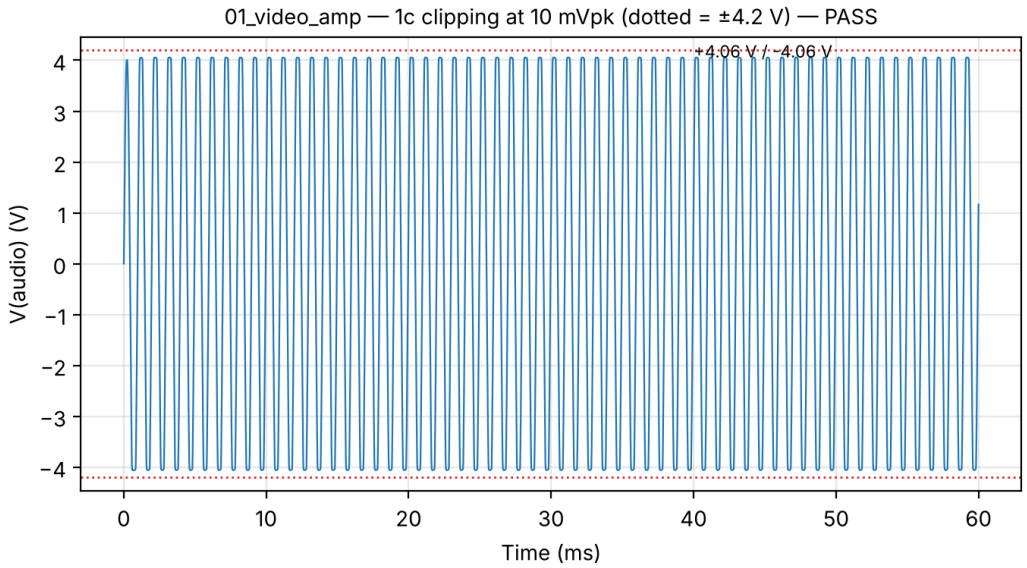



Figure 19: Clipping at 10 mV in

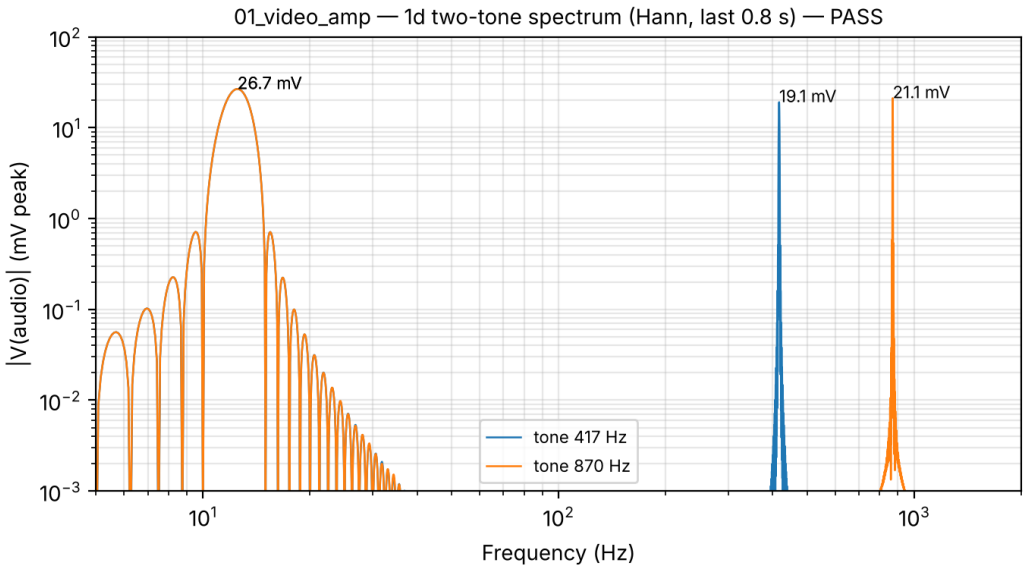



Figure 20: Two-tone spectrum (leakage + echo)

## Card 2: half-rail reference

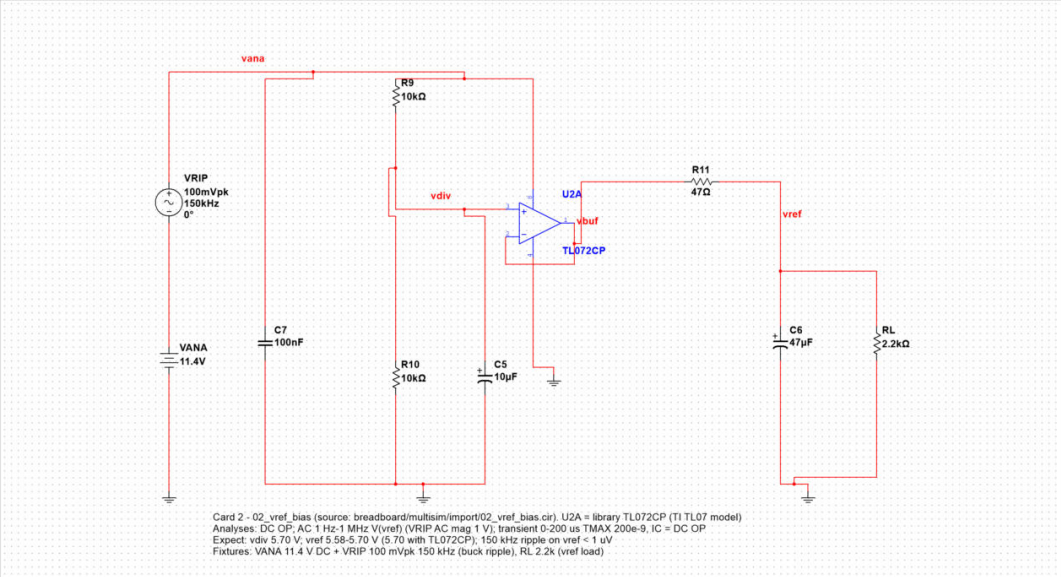



Figure 21: Multisim schematic

02_vref_bias — DC operating point — PASS

| node | DC operating point |
|---|---|
| V(vana) | 11.40000 V |
| V(vbuf) | 5.70001 V |
| V(vdc) | 11.40000 V |
| V(vdiv) | 5.70000 V |
| V(vref) | 5.58078 V |

Figure 22: DC operating point

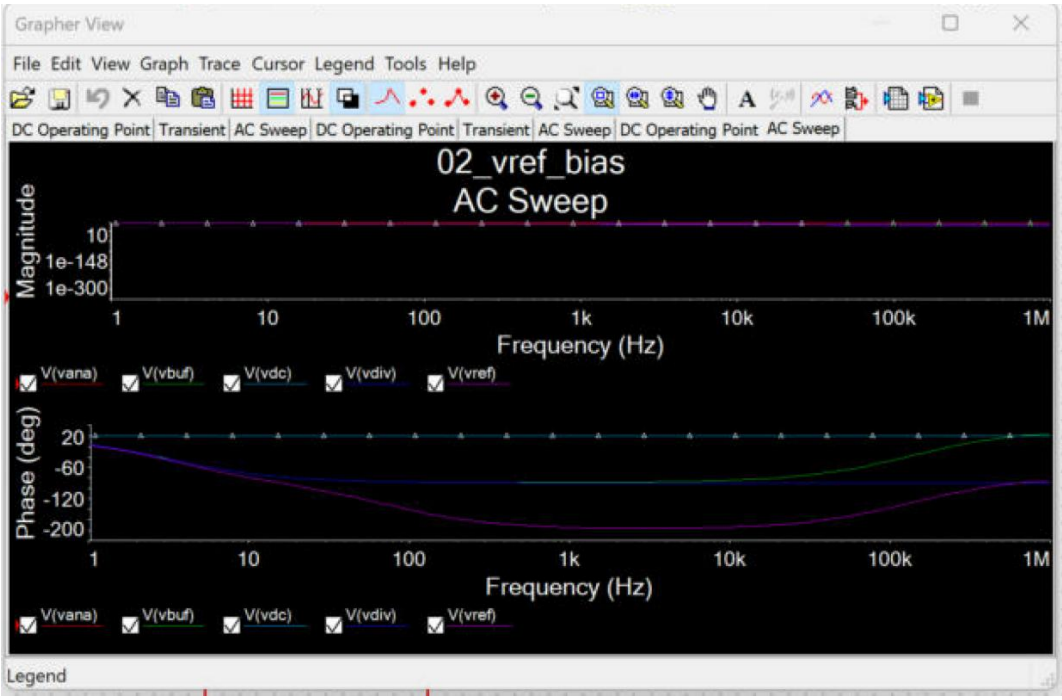



Figure 23: Multisim grapher, AC sweep

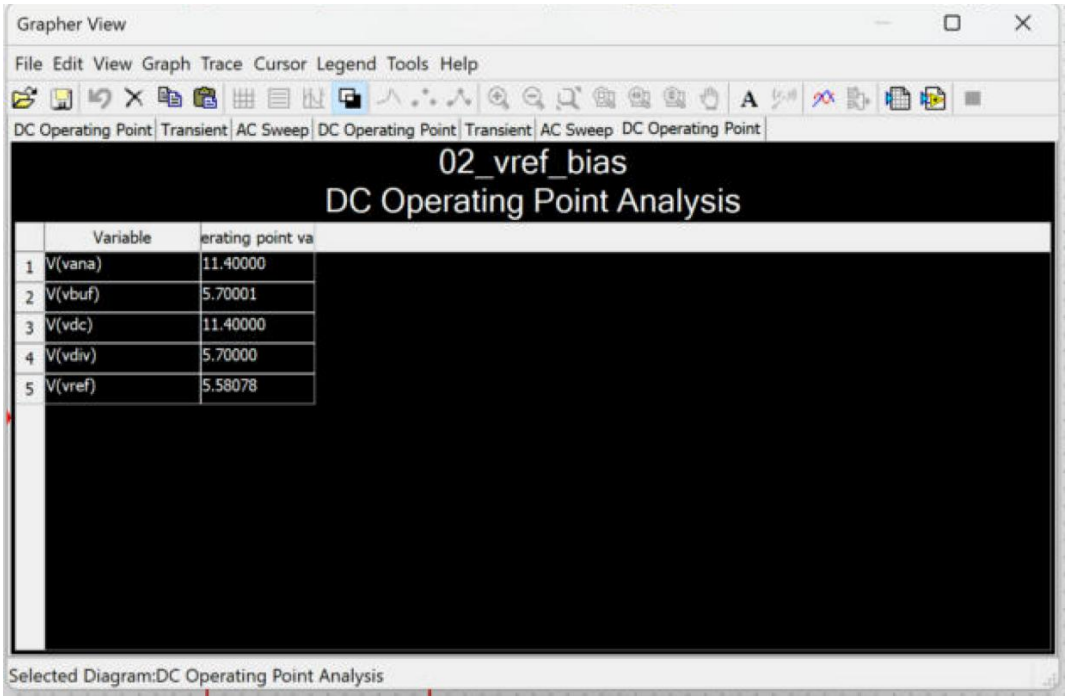



Figure 24: Multisim grapher, DC operating point

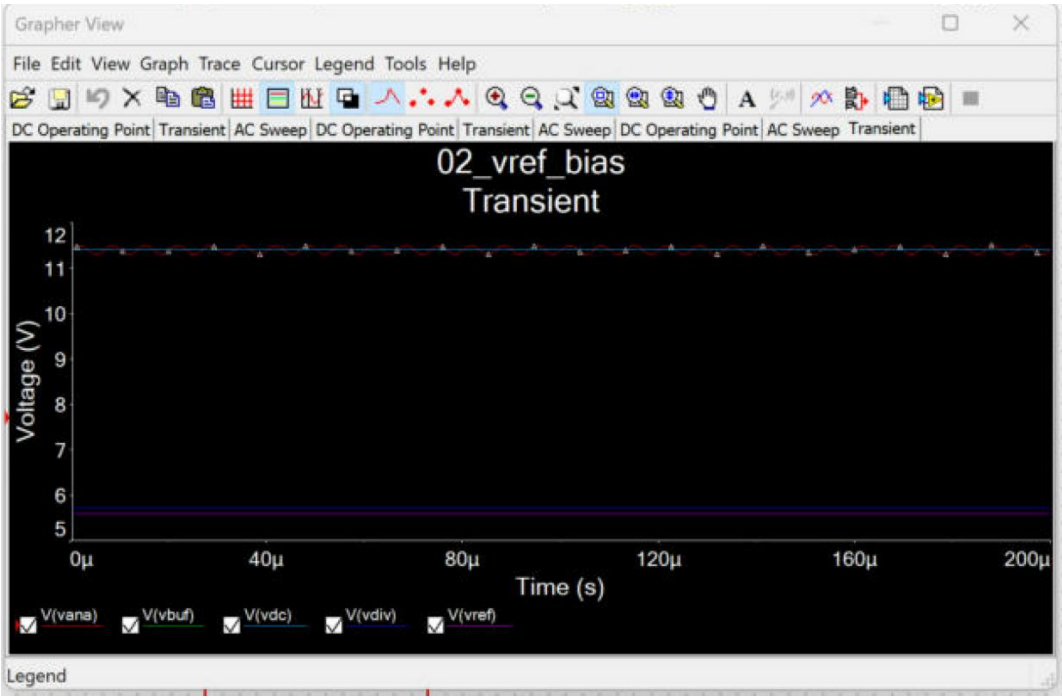



Figure 25: Multisim grapher, transient

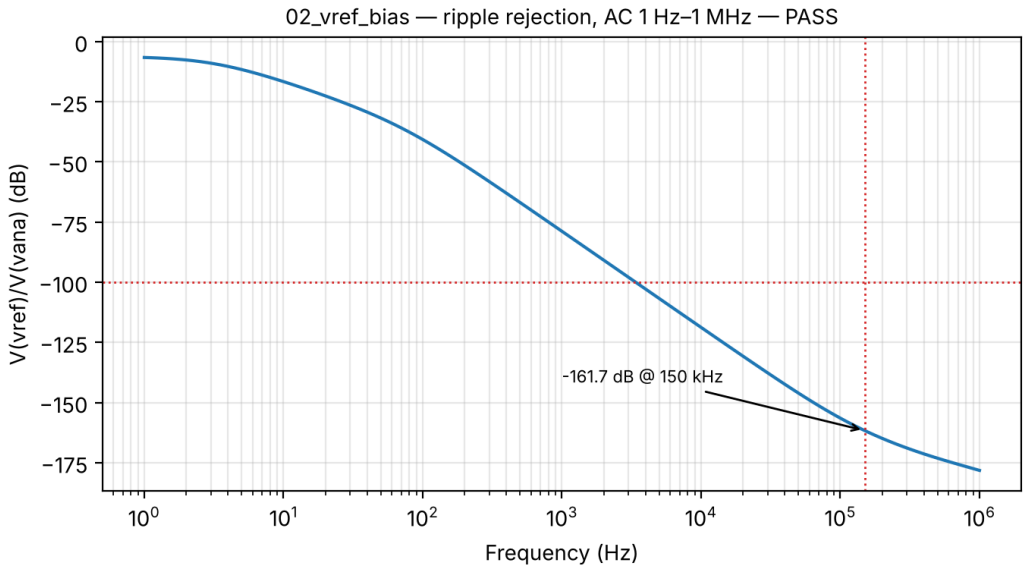



Figure 26: 150 kHz rejection to vref

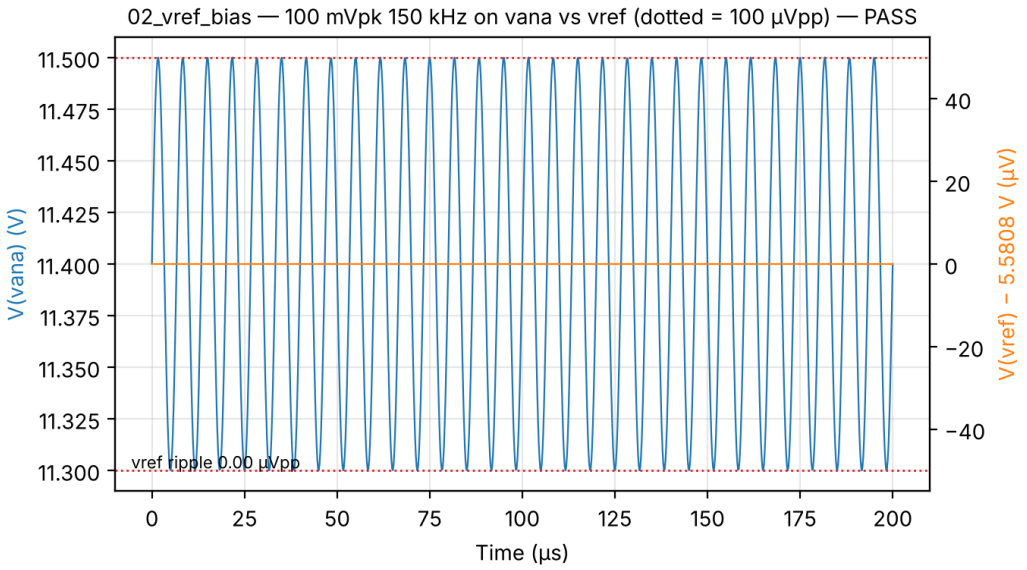



Figure 27: Ripple on vref

### Card 3: 12 V input

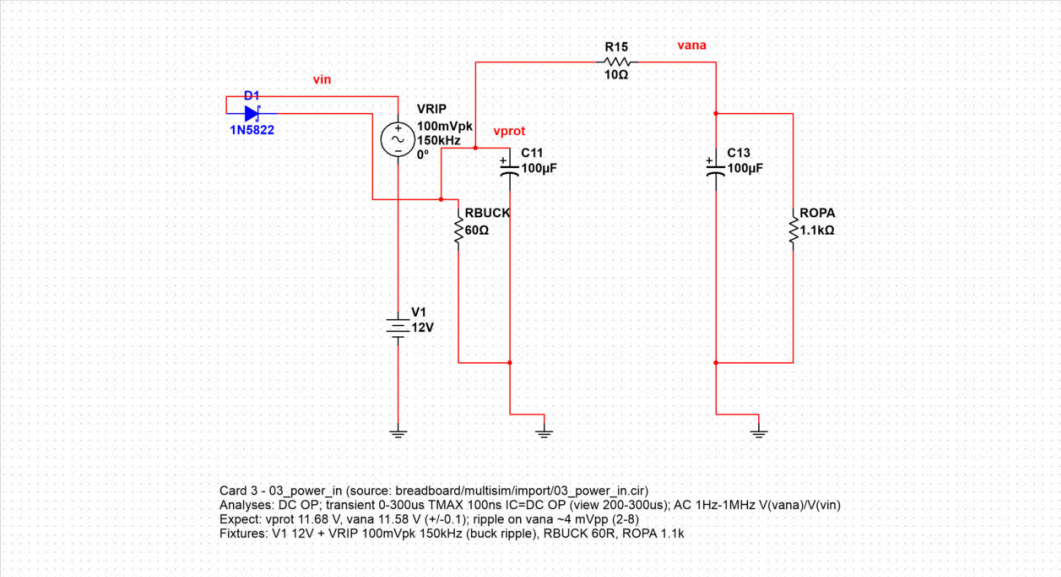



Figure 28: Multisim schematic

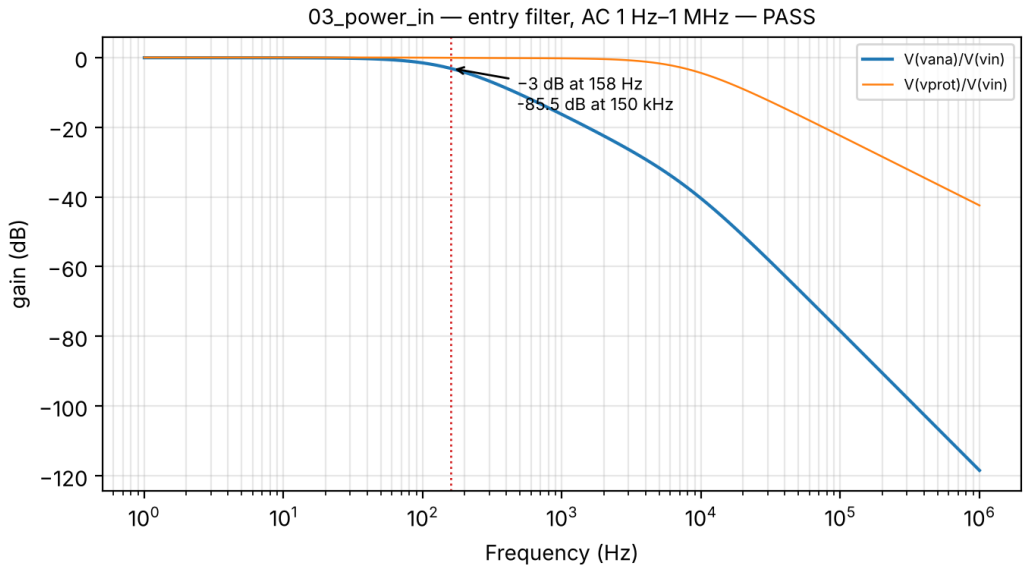



Figure 29: Input filter response

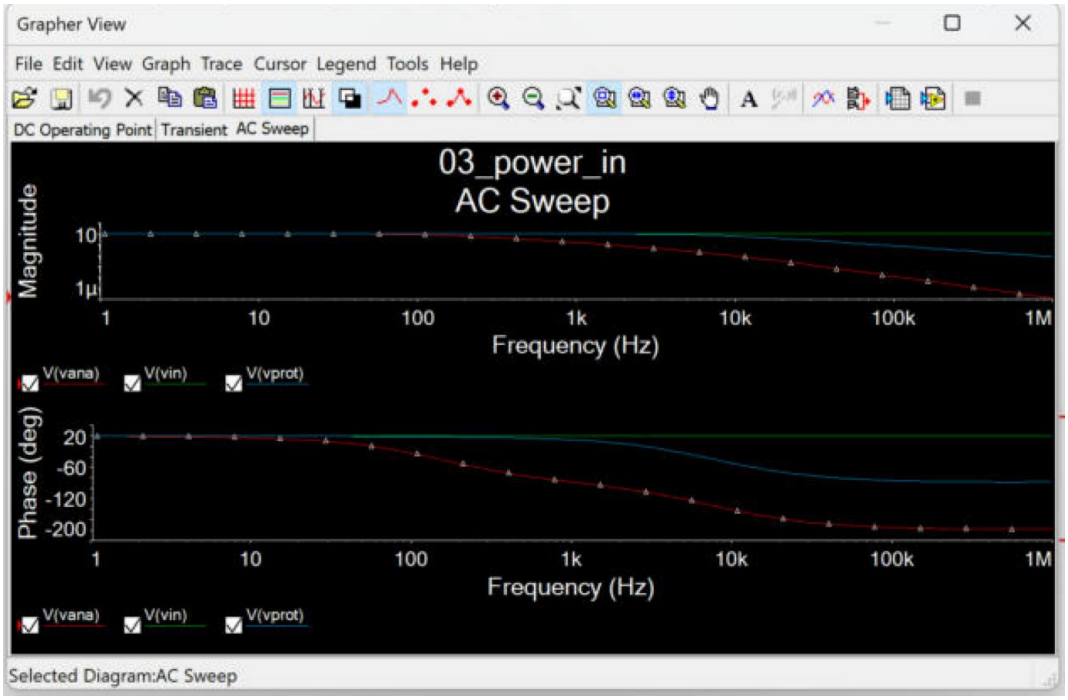



Figure 30: Multisim grapher, AC sweep

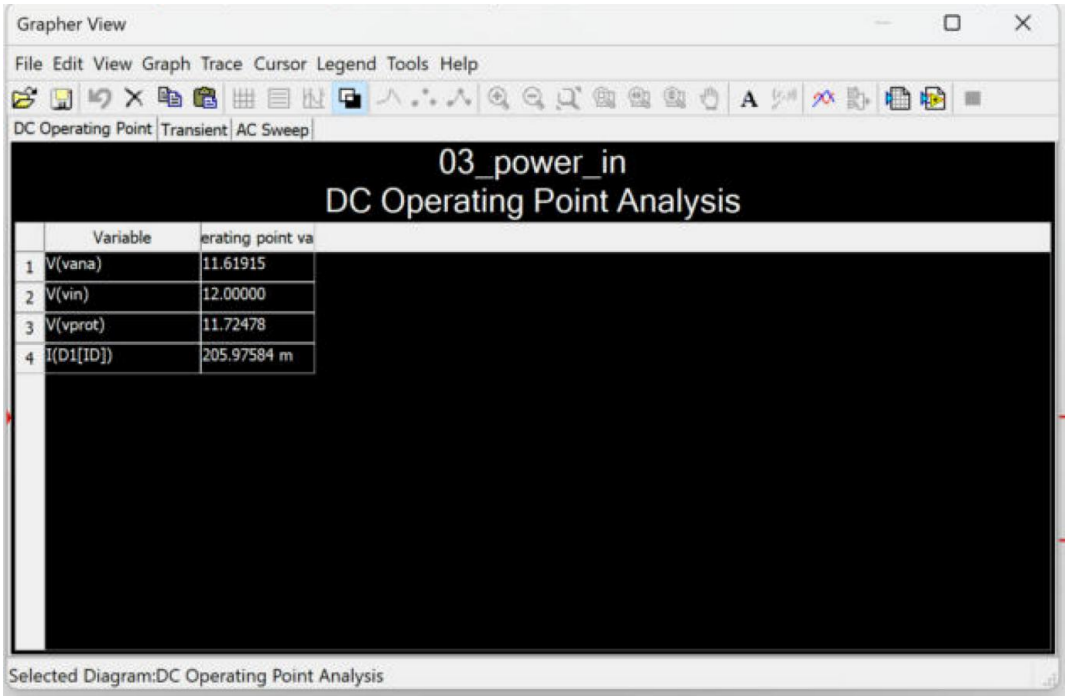



Figure 31: Multisim grapher, DC operating point

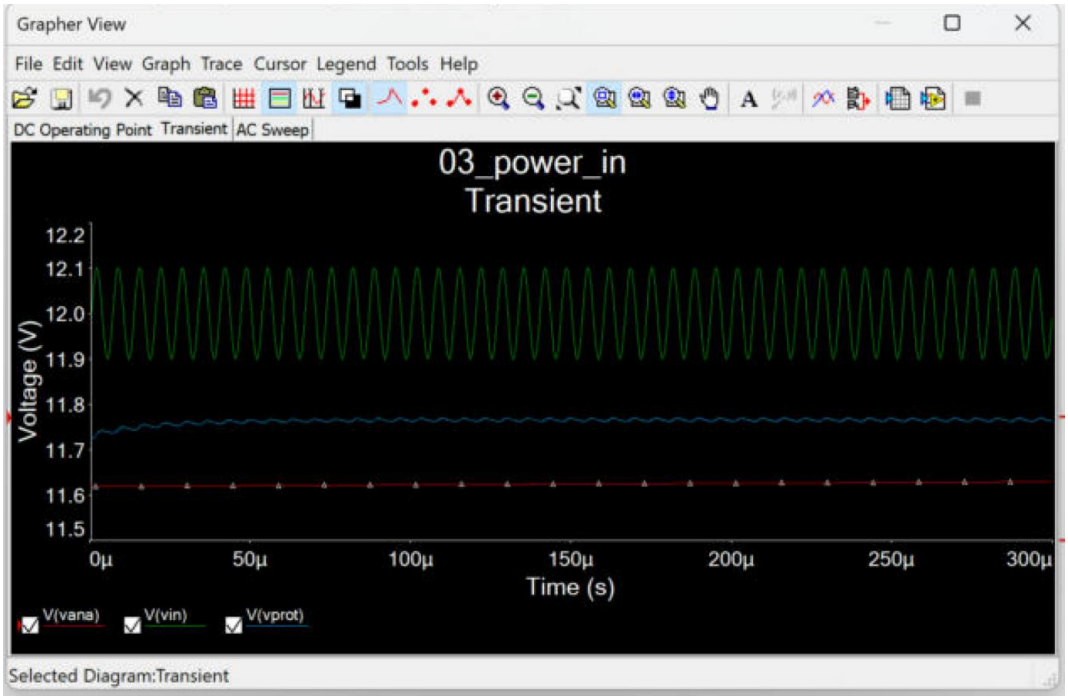



Figure 32: Multisim grapher, transient

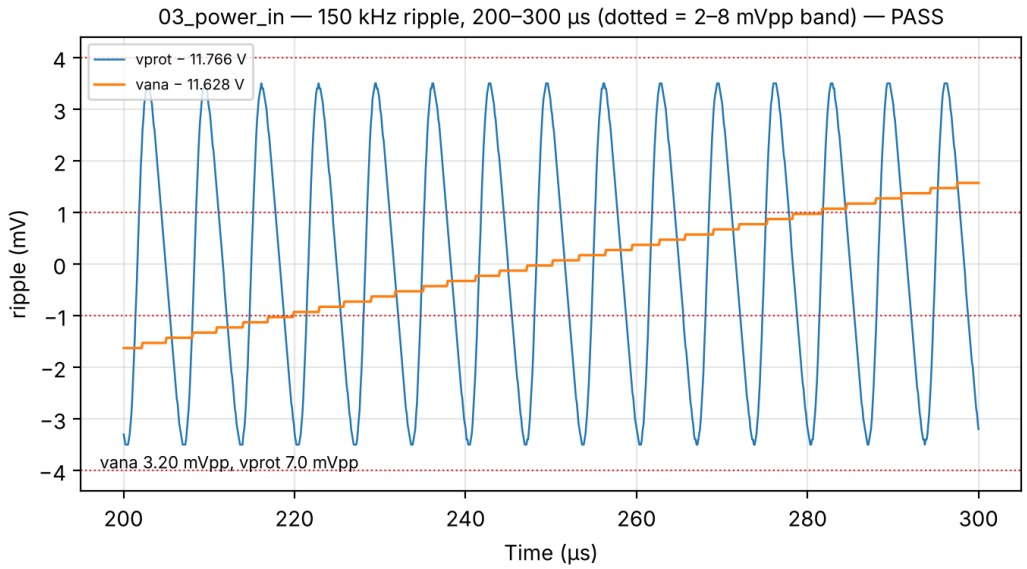



Figure 33: Ripple on the analog rail

### Card 4: reverse supply

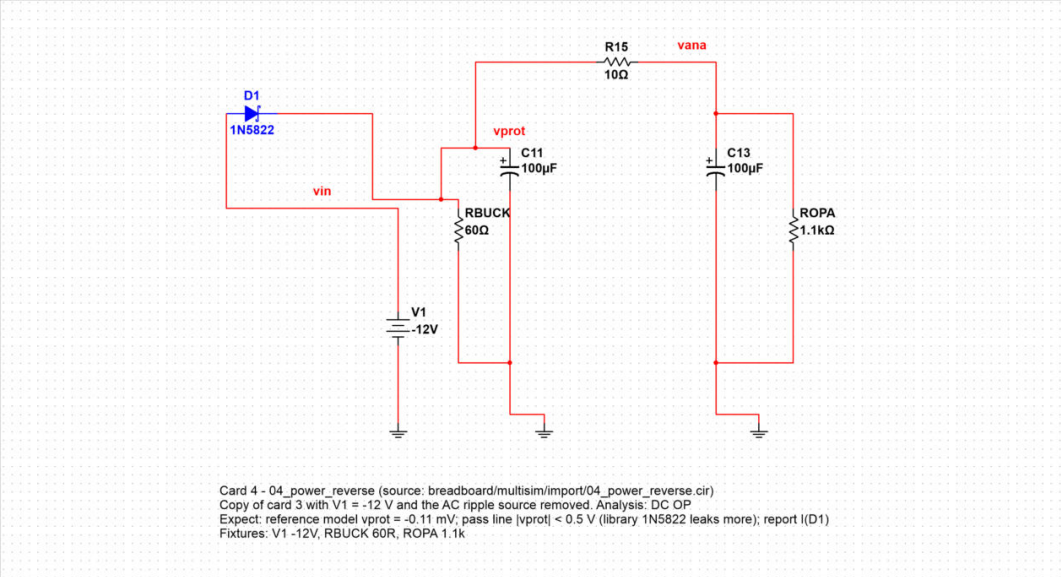



Figure 34: Multisim schematic

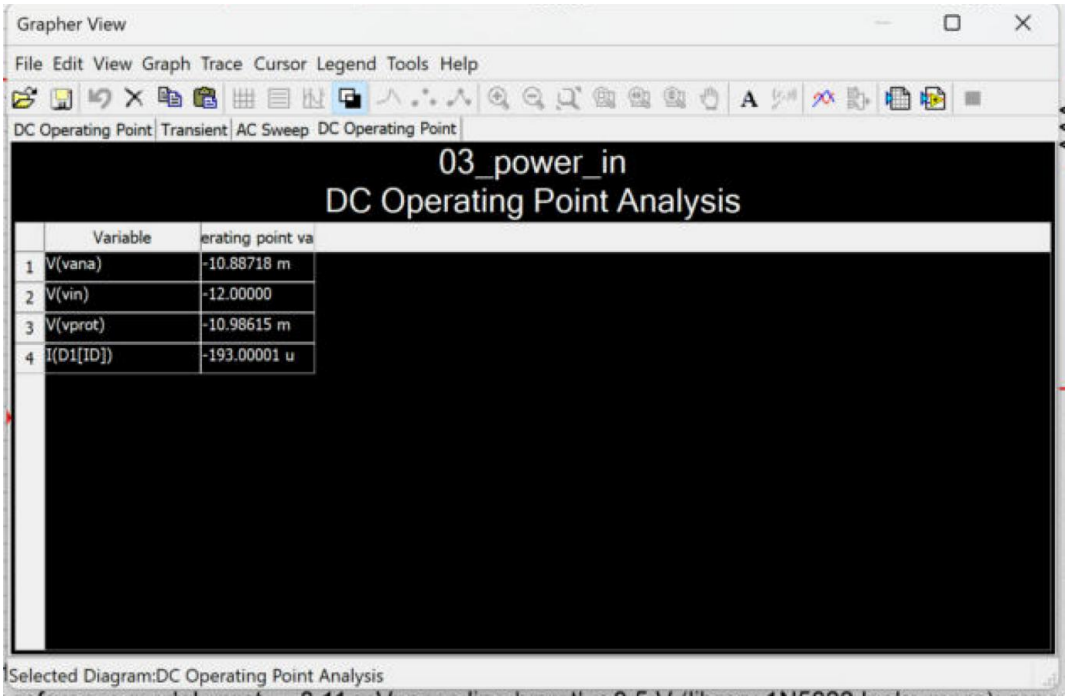



Figure 35: Multisim grapher, DC operating point

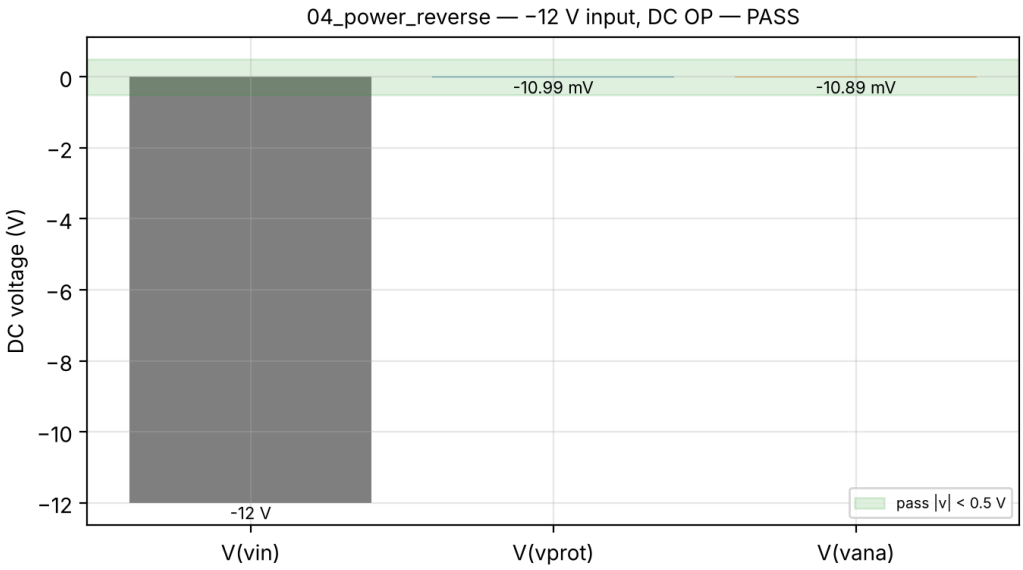



Figure 36: Reverse supply DC

**Card 5: sync divider**

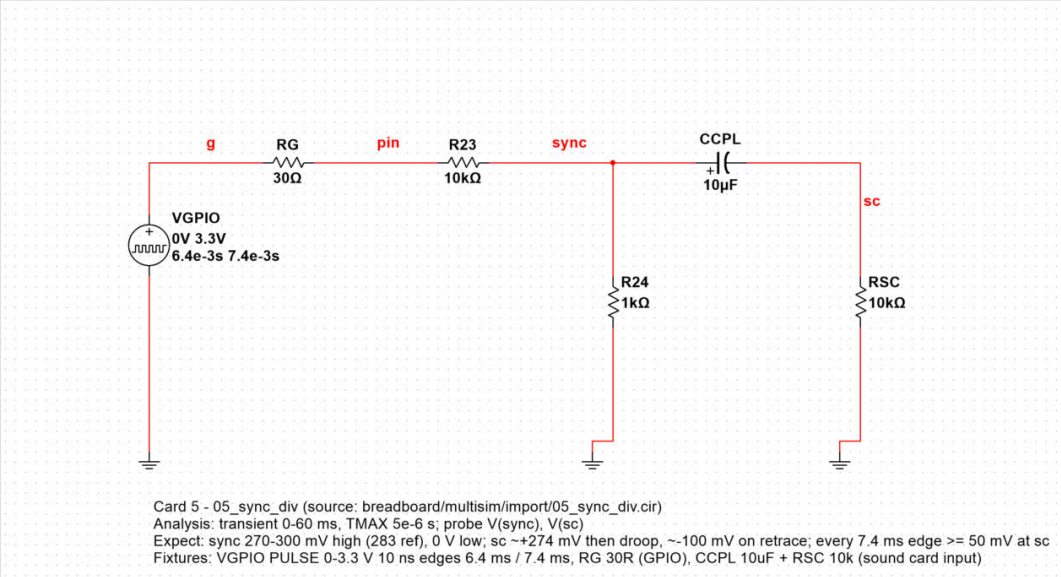



Figure 37: Multisim schematic

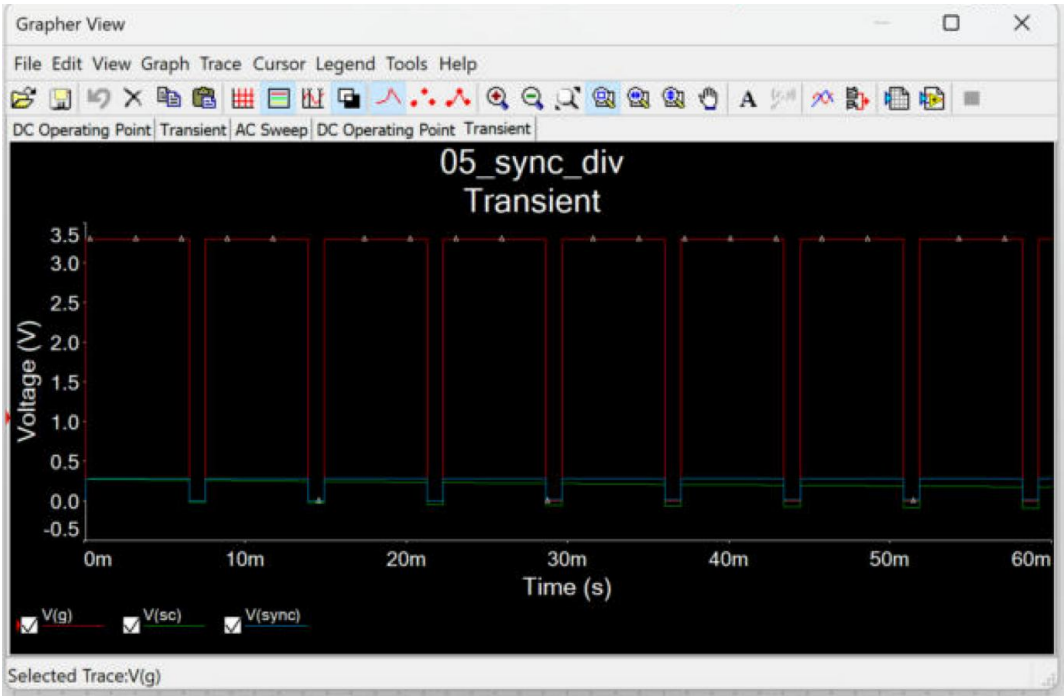



Figure 38: Multisim grapher, transient

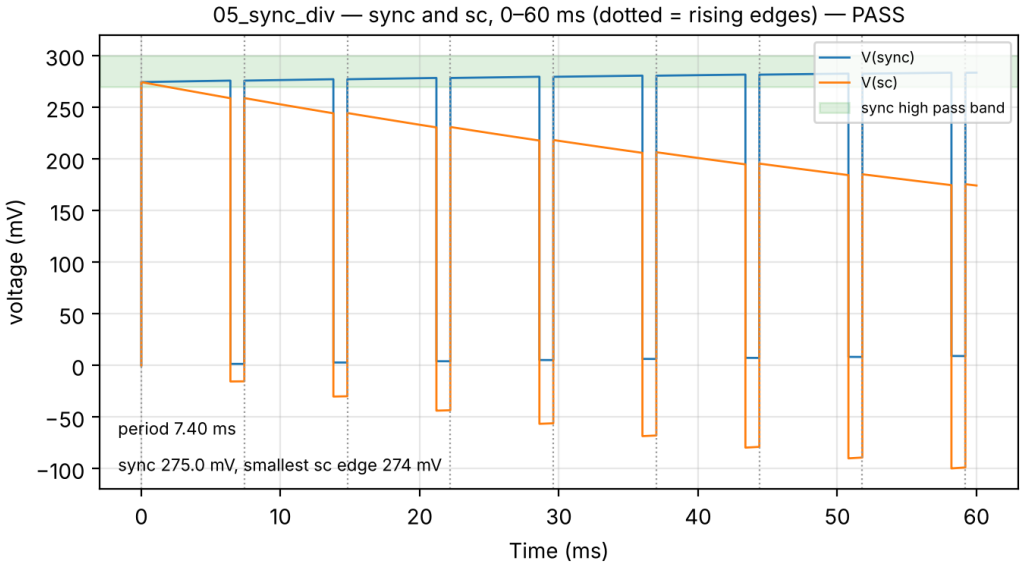



Figure 39: Sync pulse into the sound card

**Card 6: IF input**

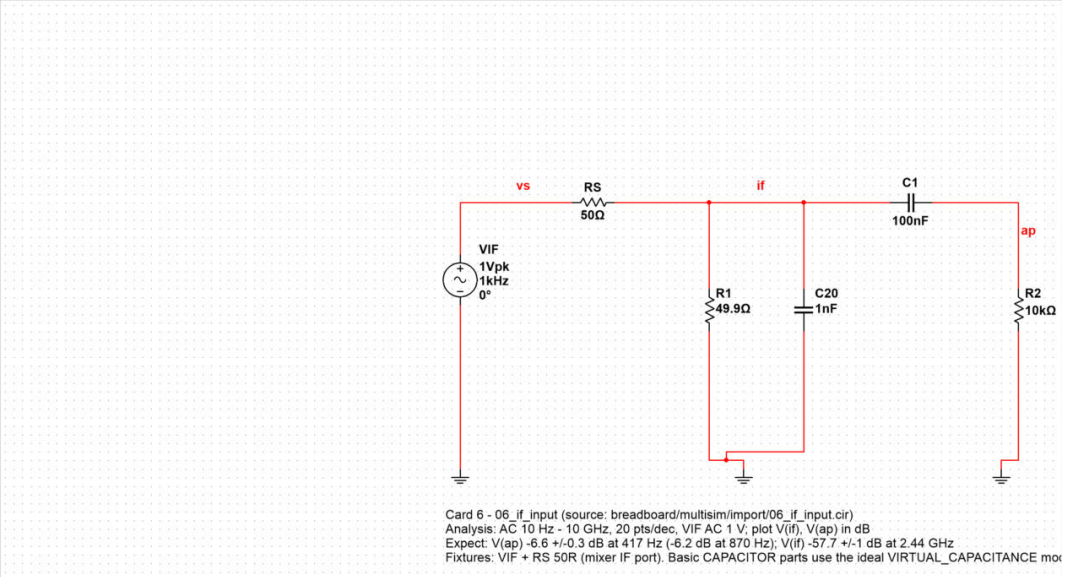



Figure 40: Multisim schematic

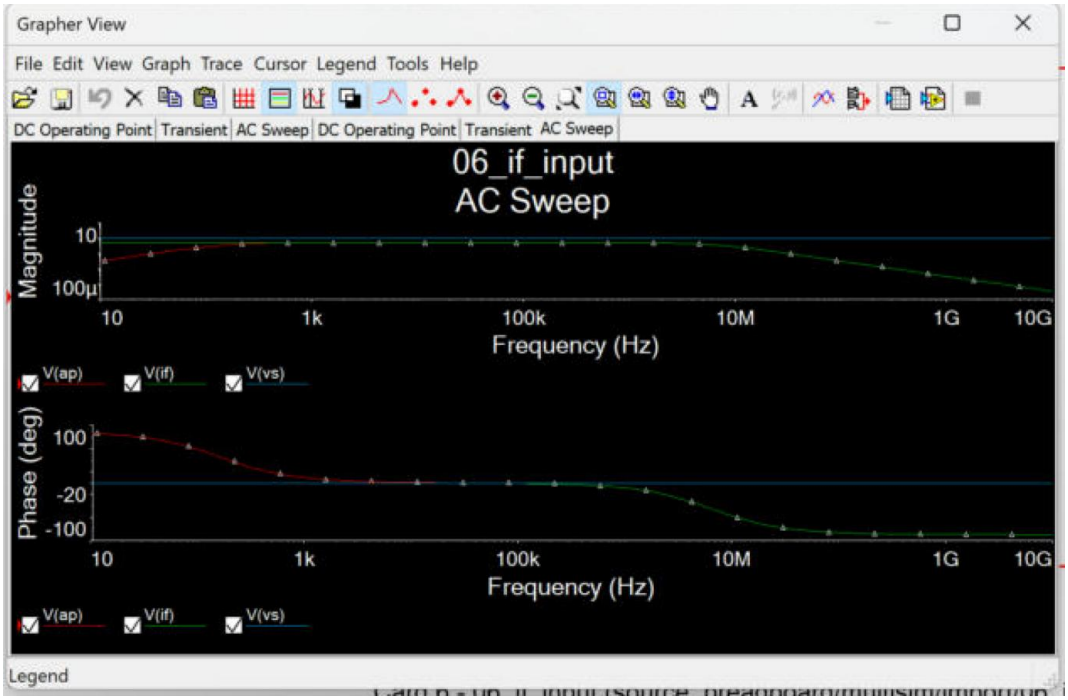



Figure 41: Multisim grapher, AC sweep

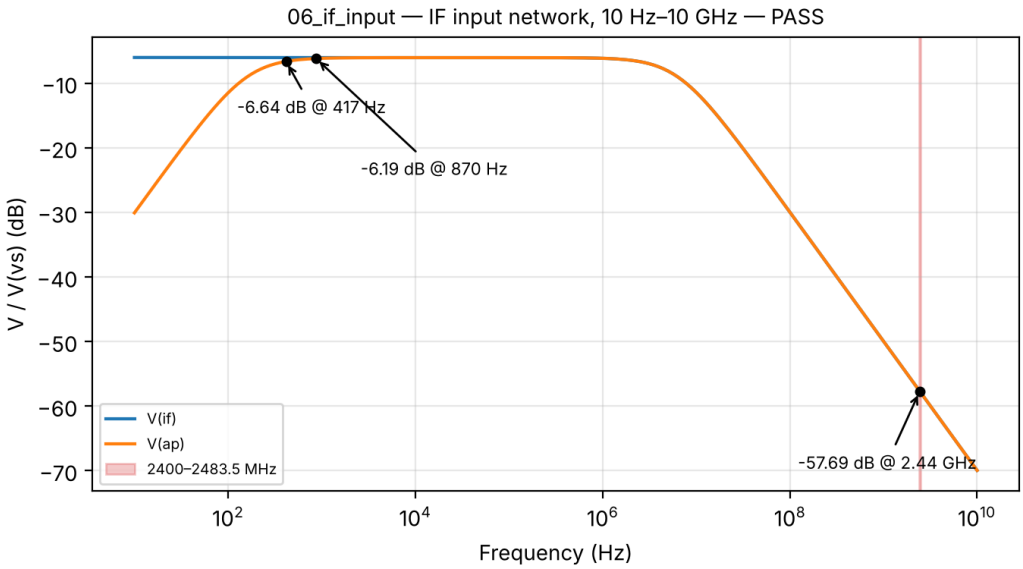



Figure 42: IF input response

**Card 7: scaled radar**

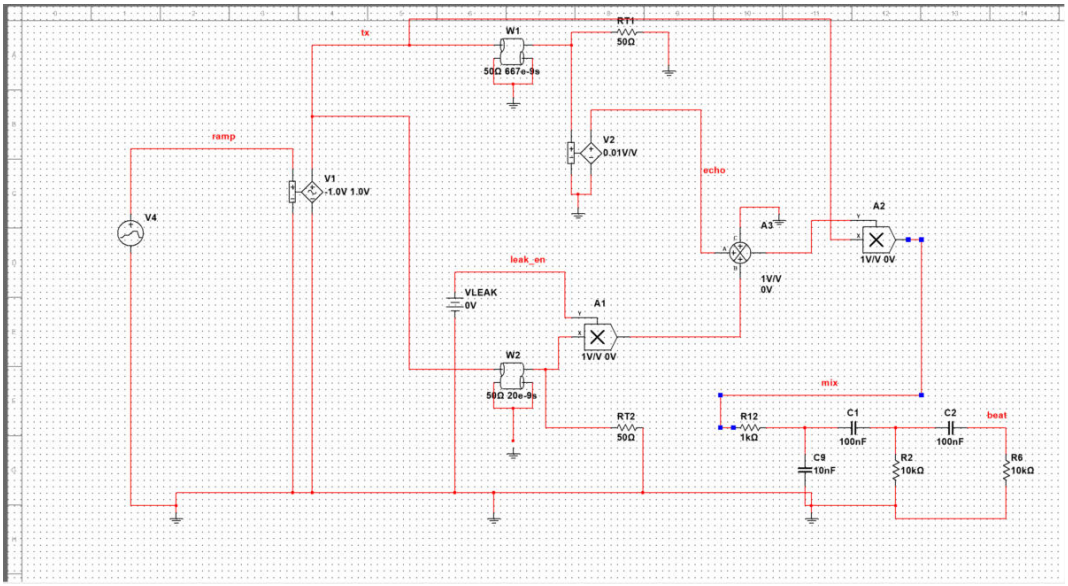



Figure 43: Multisim schematic

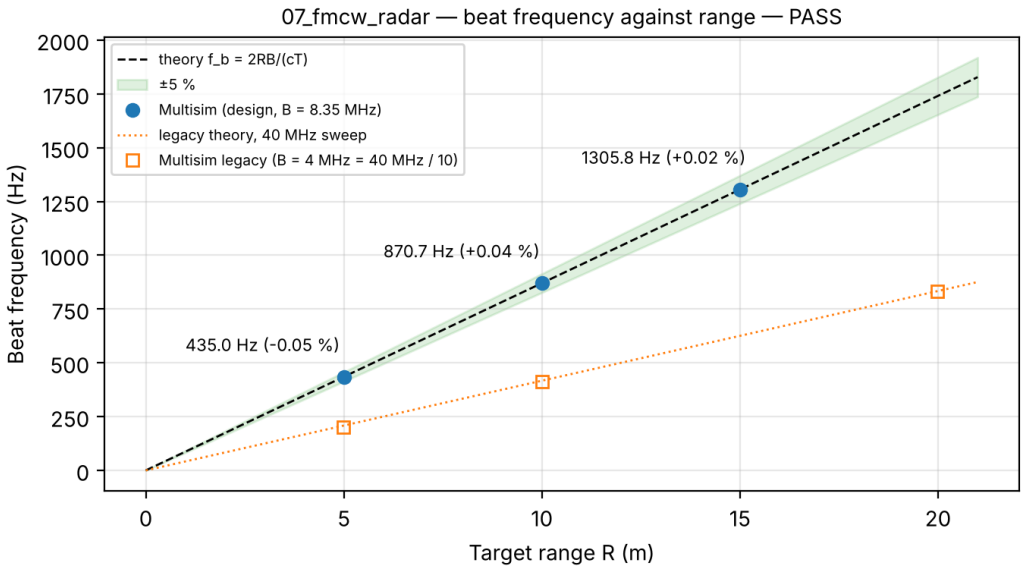



Figure 44: Beat frequency vs. range

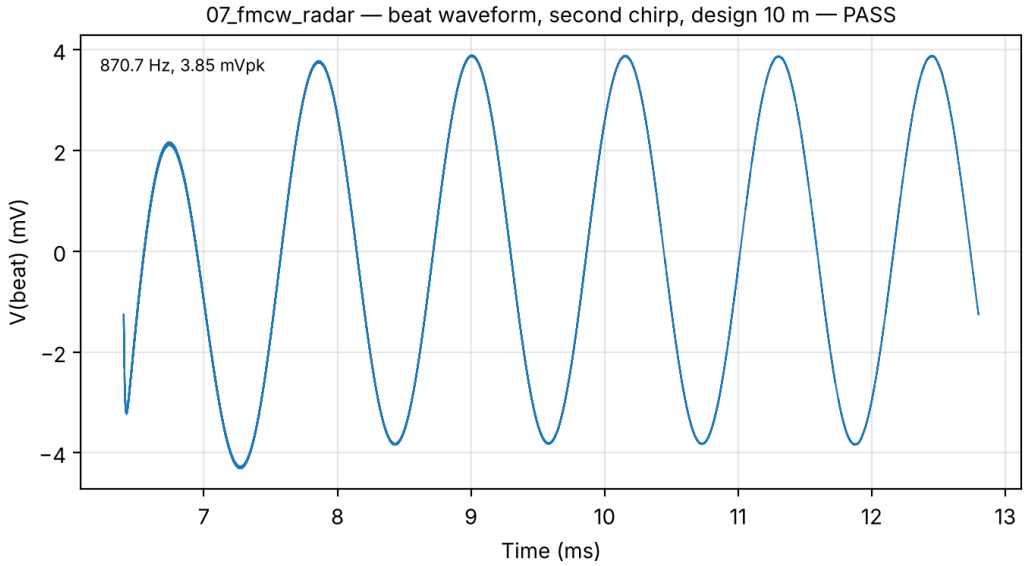



Figure 45: Beat waveform

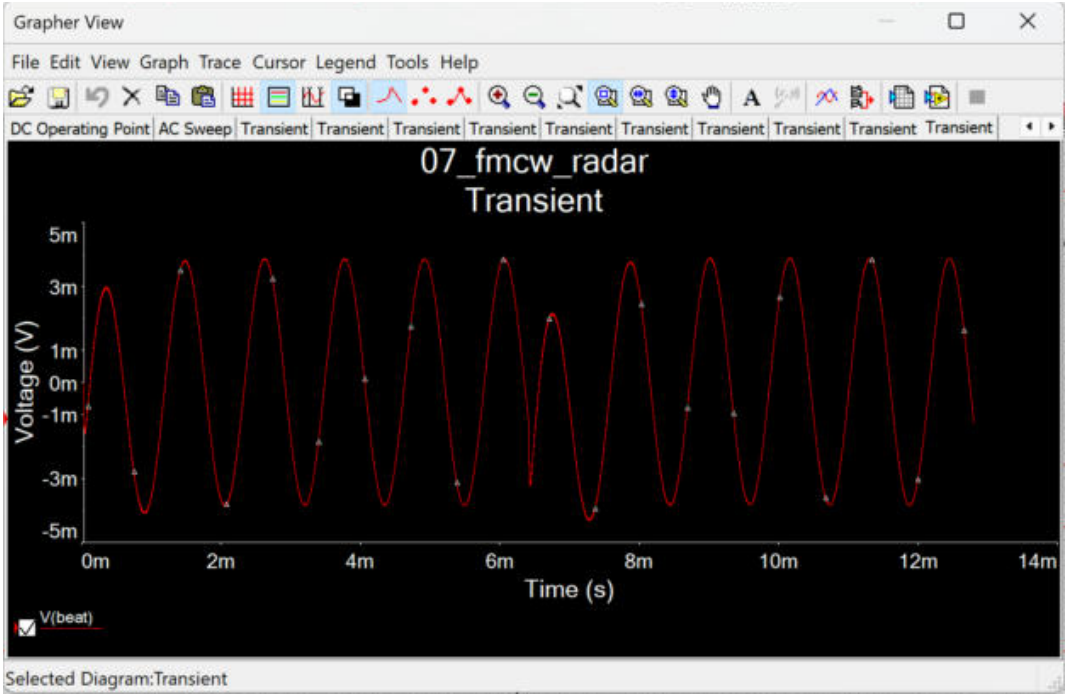



Figure 46: Multisim grapher, transient (10 m)

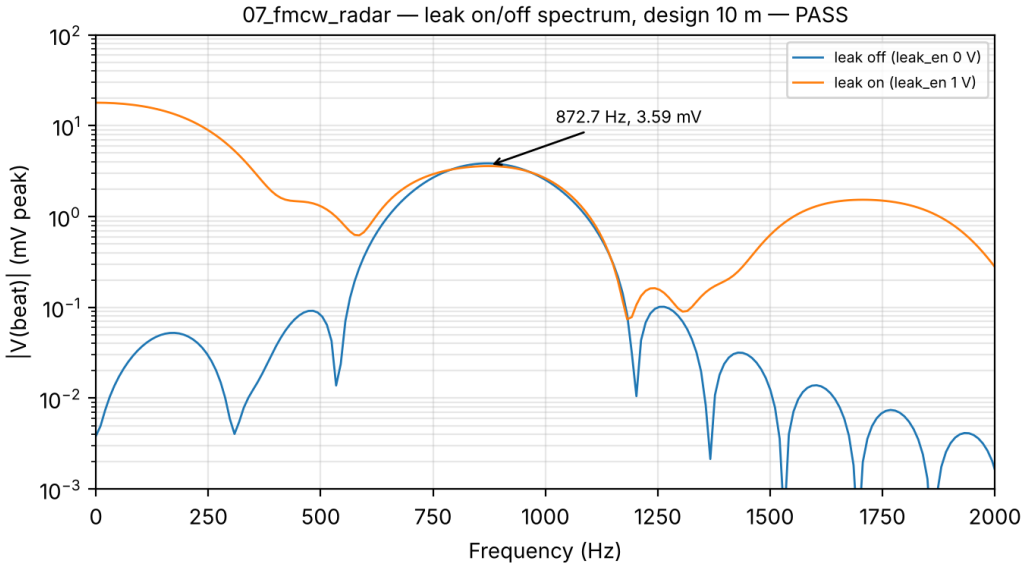



Figure 47: Spectrum with leakage on/off

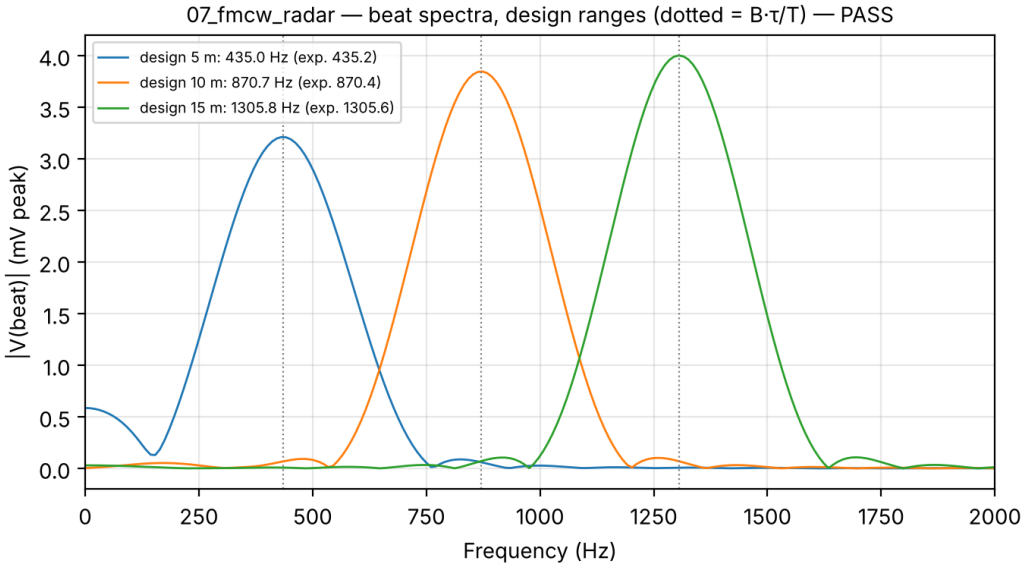



Figure 48: Spectra at each range

## Card 8: full board

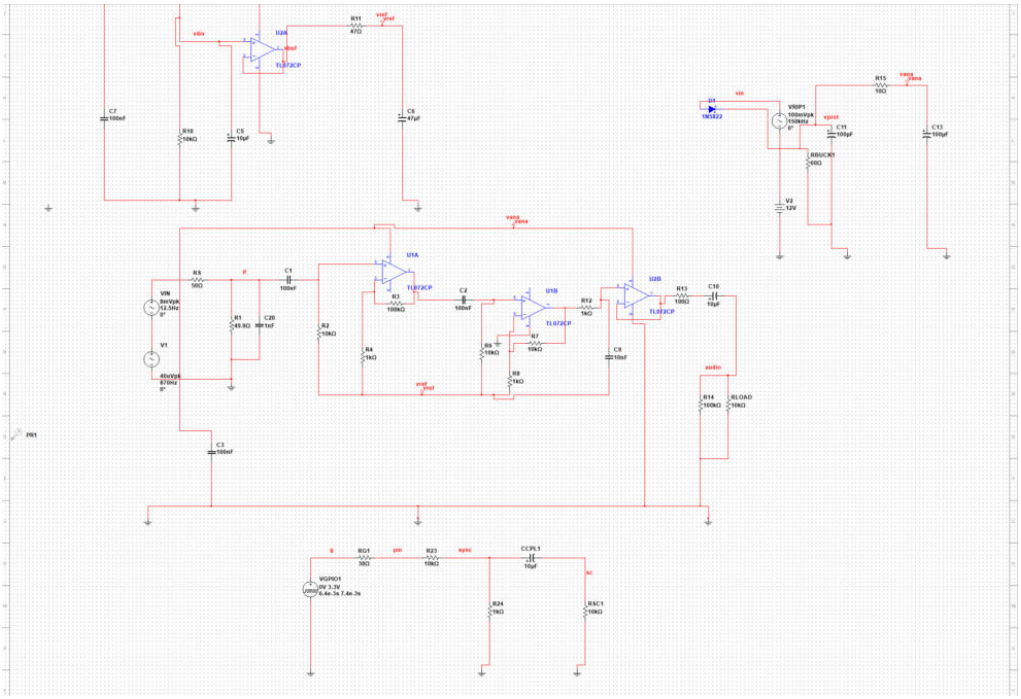


Figure 49: Multisim schematic (screenshot from before the supply-pin fix)

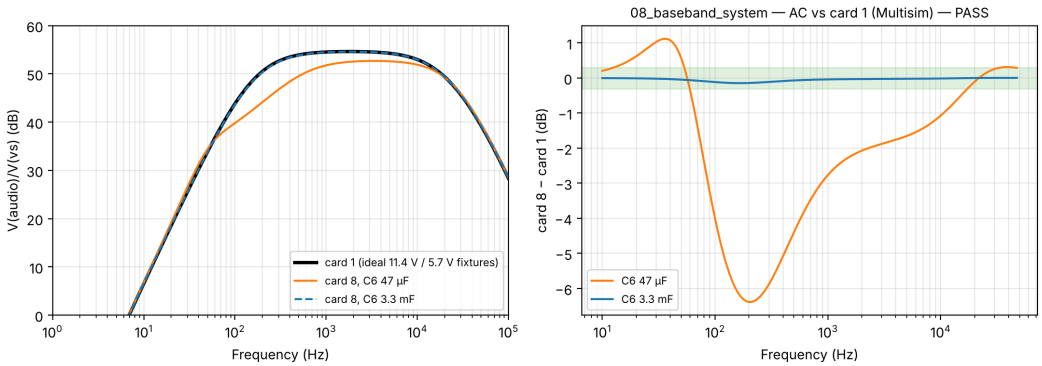



Figure 50: Gain vs. card 1, C6 47 µF and 3.3 mF

08_baseband_system — DC operating point, fixed sheet — PASS

| node | DC operating point |
|---|---|
| V(am) | 5.74711 V |
| V(aout) | 5.74818 V |
| V(ap) | 5.74710 V |
| V(audio) | 0.00000 V |
| V(bout) | 5.74722 V |
| V(bp) | 5.74710 V |
| V(if) | 0.00000 V |
| V(lp) | 5.74722 V |
| V(obuf) | 5.74723 V |
| V(pin) | 0.00000 V |
| V(sc) | 0.00000 V |
| V(sync) | 0.00000 V |
| V(vana) | 11.49418 V |
| V(vbuf) | 5.74710 V |
| V(vdiv) | 5.74709 V |
| V(vin) | 12.00000 V |
| V(vprot) | 11.72227 V |
| V(vref) | 5.74710 V |

Figure 51: DC operating point, full board (fixed)

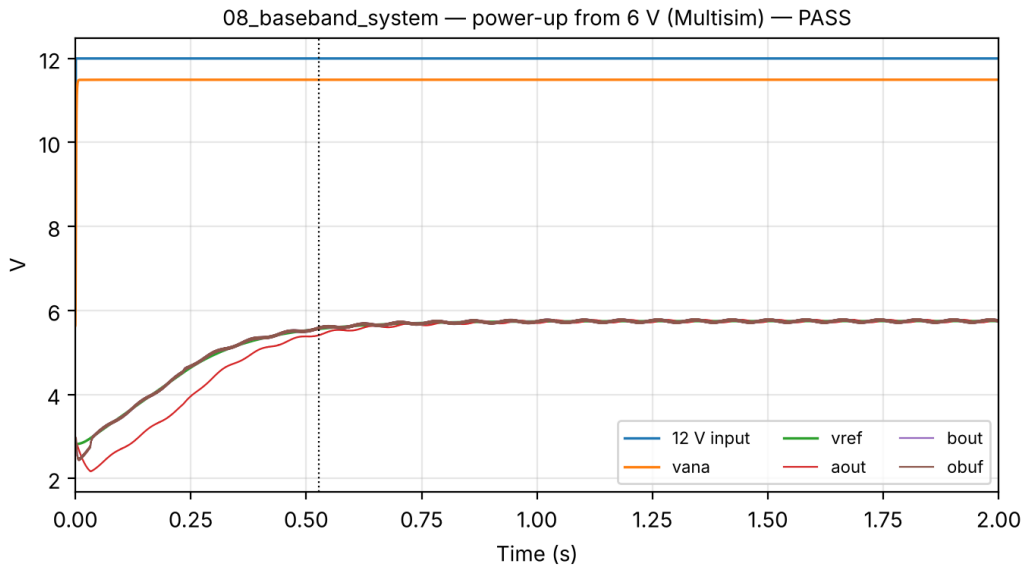



Figure 52: Power-up from 6 V

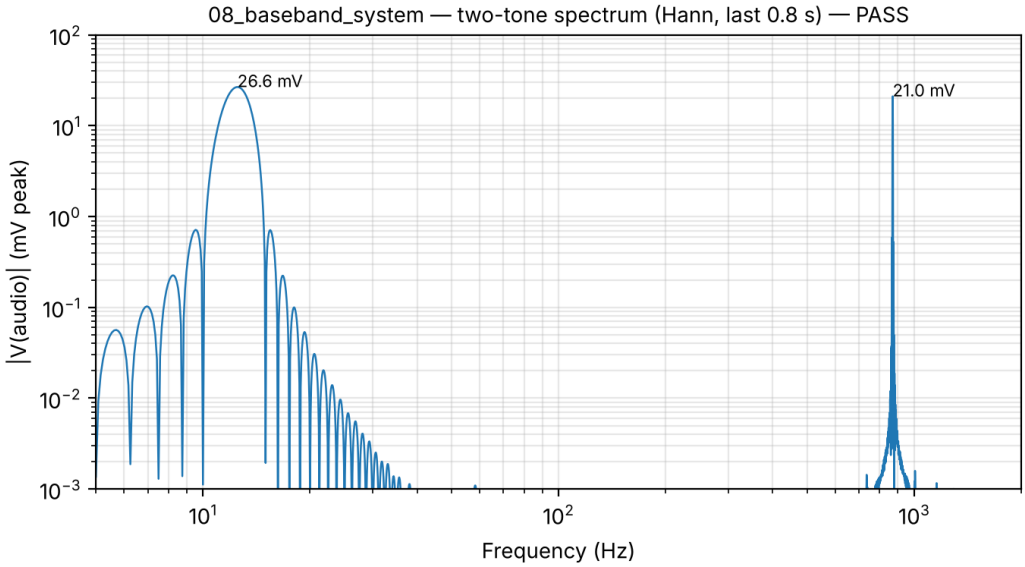



Figure 53: Two-tone spectrum, full board